

Motivation

When retrieval and task-specific context are withheld, does multi-agent decomposition improve entity resolution over a single call to the same base LLM?

Why ER matters

Entity resolution decides when separate records refer to the same real-world entity. It supports company database integration after acquisitions, census deduplication, publication disambiguation, and product-catalog matching. Missed links and false links propagate into downstream analytics, so the quality of integration and analysis is bounded by the quality of the match decisions.

Traditional burden

Rule-based systems require experts to select similarity functions and tune thresholds. Supervised and fine-tuned matchers require labeled pairs, feature design, model training, and dataset-specific validation. These methods can be strong, but adapting them to a new schema or domain introduces engineering and annotation work before a usable matcher is available.

LLM opportunity

LLMs offer a lower-setup alternative: give the model both records' attributes and request a structured match verdict. In this experiment, the LLM configurations use no task-specific training. Recent ER benchmarks suggest such matchers can be competitive; here, their Abt-Buy F1 values sit in the range DITTO [3] reports on this benchmark after fine-tuning (F1 0.893).

What I tested

Prior multi-agent ER systems often pair decomposition with retrieval or added inference context. I tested a restricted comparison using the same base LLM in both LLM arms, five local tools only in the multi-agent arm, sequential calls, and no retrieval or demonstrations. The question is whether this bundled multi-agent configuration improves over one direct LLM call, and at what cost.

Method

Same product-pair input: name, description, and price for both records.

Magellan-style RF

Attributes → 14 similarities → Trained RF → Verdict

Fourteen manually engineered similarities summarize names, descriptions, and prices. Hyperparameters and the decision threshold are selected on validation data; the final Random Forest is fitted on training plus validation pairs before test evaluation. This is a manual adaptation, not output from the Magellan library.

Single LLM

Attributes → One prompt → Verdict

One gpt-oss-120b prompt receives both records' names, descriptions, and prices and returns one structured verdict. No dataset-specific training or retrieved examples are used.

Multi-agent

Attributes → Two reviewers → Orchestrator → Verdict

Syntactic and semantic reviewers use the same gpt-oss-120b base model. Their structured outputs go to a same-model orchestrator, which returns the final verdict. Reviewer calls are sequential.

Structured response

```
{
  "verdict": "MATCH" | "NO MATCH",
  "confidence": 0.0 to 1.0,
  "reasoning": "short justification"
}
```

MATCH or NO MATCH is the evaluated verdict; confidence and reasoning are recorded, never thresholded.

RF feature set (14)

Name (8): Jaccard q3-gram, cosine tokens, Jaccard tokens, Monge-Elkan, Levenshtein distance, Levenshtein similarity, Needleman-Wunsch, Smith-Waterman
Description (2): Jaccard q3-gram, TF-IDF cosine
Price (4): exact match, normalized absolute difference, Levenshtein distance, Levenshtein similarity

Tools and restriction

Five local tools support the multi-agent reviewers: SequenceMatcher name similarity, token Jaccard, BM25, price ratio/difference checks, and MiniLM name similarity. Descriptions remain prompt inputs but are not inputs to these tools. Calls are sequential; no retrieval, demonstrations, or task-specific inference context are provided.

Experiments

1,916 test pairs

206 matches (10.8%)

Five executions per configuration

Same fixed pair set

Experiment procedure

Fixed pairs → Structured verdicts → Precision / recall / F1 → Execution-level test → Heuristic error analysis

Both LLM configurations use temperature zero and structured responses. Requests and multi-agent reviewer calls run sequentially; parallel execution was not tested.

Parse failures

A reply that fails schema parsing is recorded as NO MATCH at confidence 0.0 and kept in the evaluation, never dropped. Single LLM: 0 parse failures across all five runs. Multi-agent: 4, 4, 4, 5, 4 per run (21 of 9,580 pair decisions). 20 of those 21 fell on pairs labeled NO MATCH, so the default was already the correct call for all but one; dropping them shifts mean F1 by 0.0001, and the default could have flattered multi-agent by at most 0.0084 F1.

An exact execution-level unpaired studentized permutation test compares the five F1 values per LLM configuration. Entity-cluster sensitivity is reported separately because all executions reuse the same record pairs.

How errors were analyzed

Every prediction is joined to its gold label on (source id, target id). MATCH maps to 1 and NO MATCH to 0, giving TP (1/1), FP (1/0), FN (0/1), and TN (0/0). Precision = TP / (TP + FP) and recall = TP / (TP + FN) are computed from those counts. Deterministic heuristic categories are then assigned from product fields and generated rationales. Categories describe recurring evidence patterns; they do not verify decision mechanisms.

Single-LLM false positive | Abt 1060 vs. Buy 662

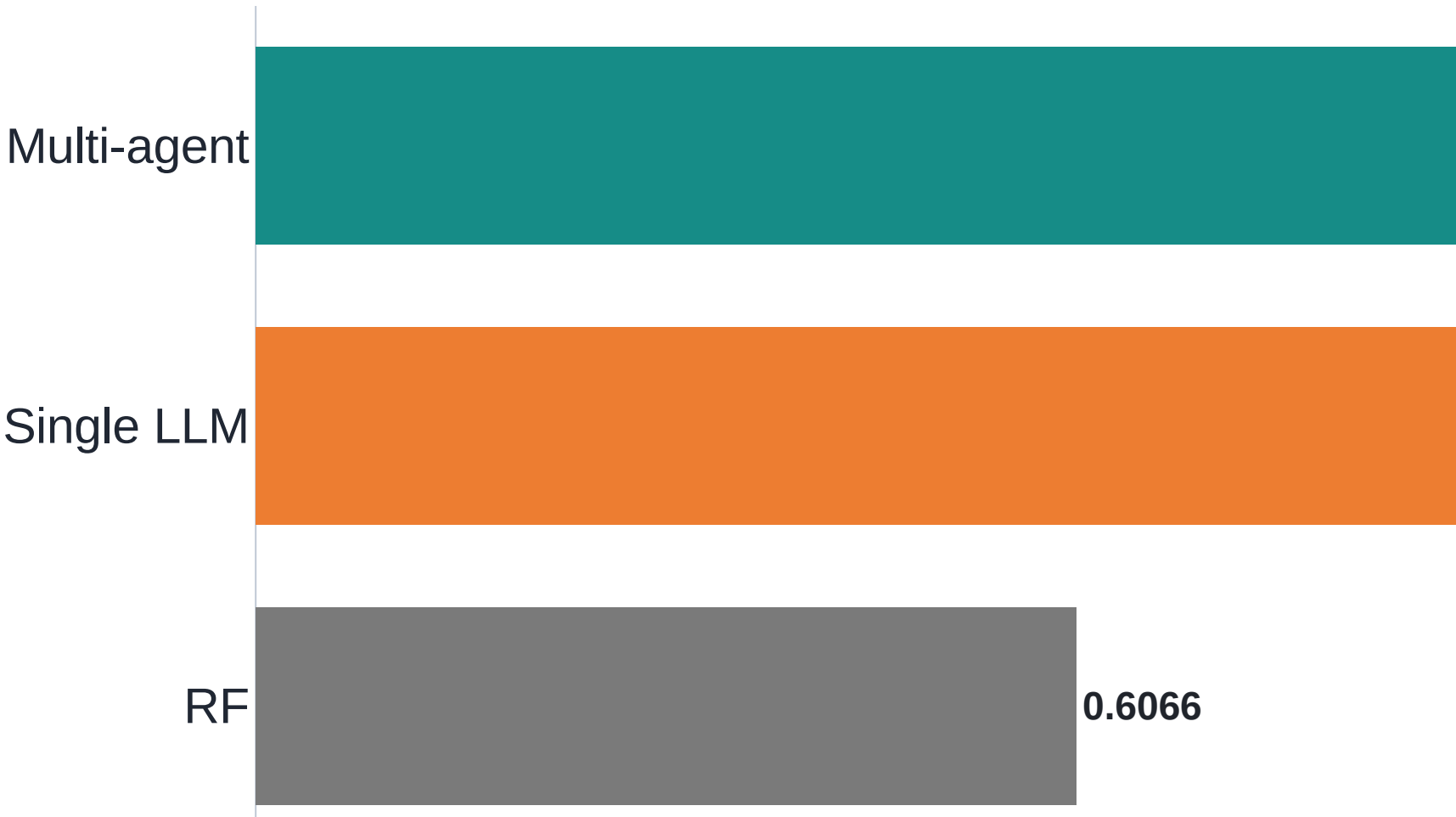
A: Canon EOS Rebel XSi, silver, 18-55 mm, 2756B003
B: Canon EOS Rebel XSi, black, 18-55 mm, 2756B003
Gold: NO MATCH | Single: MATCH (0.85)
Same base model and lens, but a different finish; the benchmark labels this pair NO MATCH.

Multi-agent false negative | Abt 121 vs. Buy 371

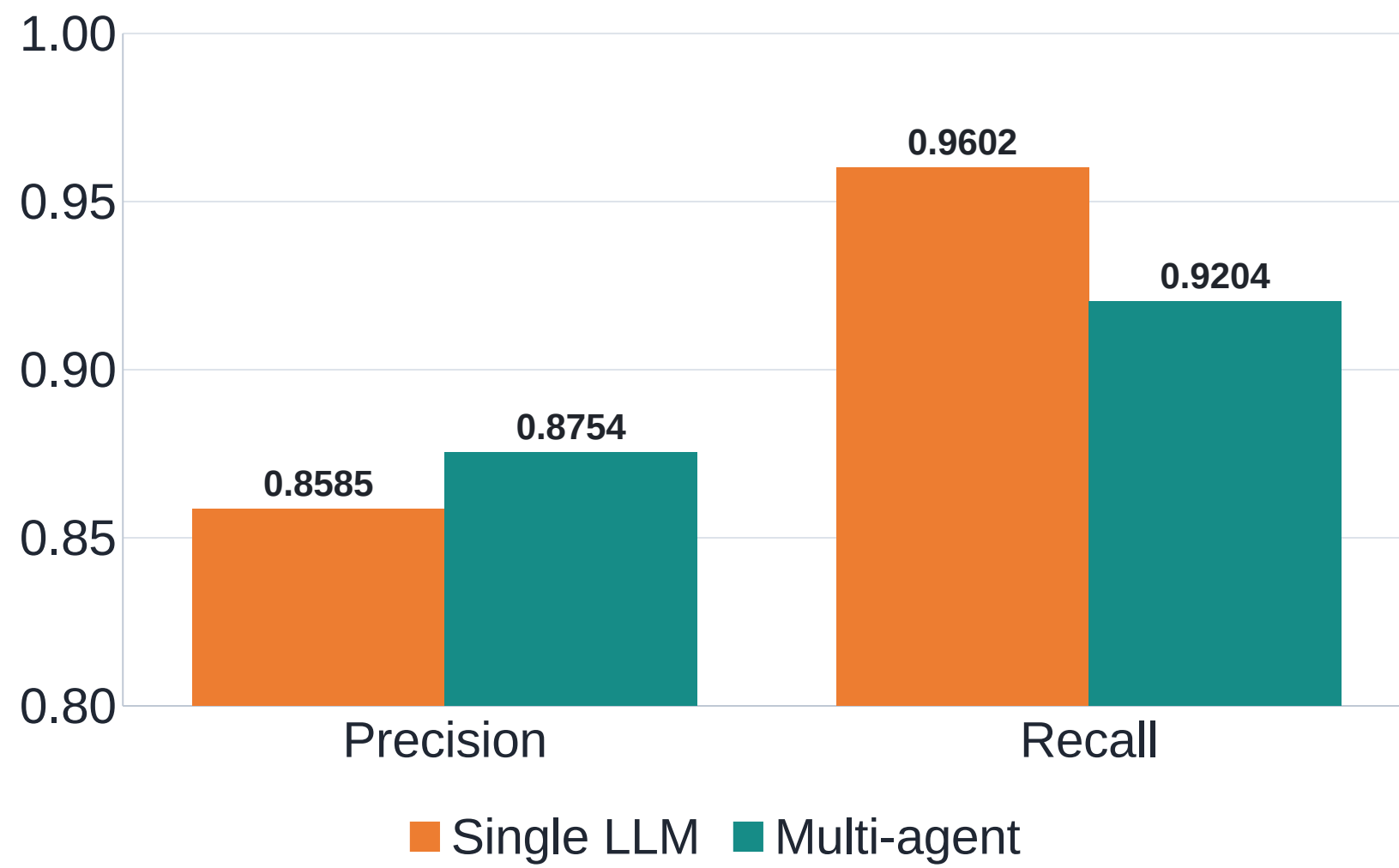
A: TiVo wireless adapter AG0100
B: Tvio wireless USB network adpator
Gold: MATCH | Multi-agent: NO MATCH (0.75)
With the Buy listing reduced to a sparse, misspelled title, the multi-agent pipeline missed a labeled match.

Results

Mean F1 (5 runs)



Precision-recall operating point



Metric	RF	Single	Multi-agent
Precision	0.6876	0.8585	0.8754
Recall	0.5427	0.9602	0.9204
F1	0.6066 ± 0.0161	0.9065 ± 0.0039	0.8973 ± 0.0048
Tokens/pair	0	656.1	5,720.0

Agreement

98.8% agreement across 25 cross-run comparisons; Cohen's kappa 0.9426 ± 0.0075.

Token and runtime cost

656 vs. 5,720 tokens/pair (8.7x); separately timed sequential runs took about 42 vs. 281 minutes (6.6x).

Illustrative errors and risk profile

Illustrative run (eval_run_0), all 1,916 pairs: Single 33 FP / 8 FN; Multi-agent 26 FP / 14 FN. The single LLM was descriptively more aggressive, retaining higher recall but making more false positives on close variants and accessories labeled as distinct. The multi-agent configuration was more conservative, reducing false positives while missing more labeled matches, especially sparse listings.

Confidence observation

Confidence signal, not calibration: illustrative single-LLM mean confidence was 0.924 on true positives and 0.764 on false negatives, a 0.161 gap. The model reports less confidence on pairs it goes on to miss, so a threshold could route uncertain pairs to review. These are self-reported scores, not validated probabilities, and no threshold was tuned here.

Statistical scope

Execution-level delta F1 = +0.0092 for single LLM, p = 0.0238: exactly 6 of the 252 possible relabelings of the ten executions give a gap this large, so p is enumerated, not estimated. It supports only the claim that the single LLM scored higher across repeated runs on this fixed pair set; it does not show multi-agent is worse in general. Entity-cluster sensitivity [-0.0226, +0.0443] allowed either ordering.

Conclusions

What I learned

- The single LLM reached mean F1 0.9065 without task-specific training, above this study's manual RF baseline and in the range DITTO reports on Abt-Buy after fine-tuning.
- The tested multi-agent configuration did not improve mean F1; it shifted to higher precision and lower recall.
- Despite 98.8% agreement, multi-agent used 8.7x more tokens and was 6.6x slower in separate sequential timing runs.
- Selective routing is a more promising next test than applying the full multi-agent pipeline to every pair.

Limitations

- One English, two-table product benchmark.
- One base model and five executions.
- Bundled configuration: prompts, tools, call count, reviewer outputs, and orchestration changed together.
- No retrieval or heterogeneous-agent comparison.
- Sequential rather than parallel execution.
- Heuristic error categories and uncalibrated confidence.

Future work

- Test multiple datasets and domains.
- Ablate retrieval and local tools.
- Compare heterogeneous agents and parallel reviewers.
- Use held-out calibration for confidence-guided routing to rules or human review.

References

- Binette and Steorts, 2022. (Almost) all of entity resolution.
- Konda et al., 2016, Magellan.

- Li et al., 2020, Deep entity matching with pre-trained language models.
- Althaf et al., 2025, Multi-Agent RAG Framework for Entity Resolution.